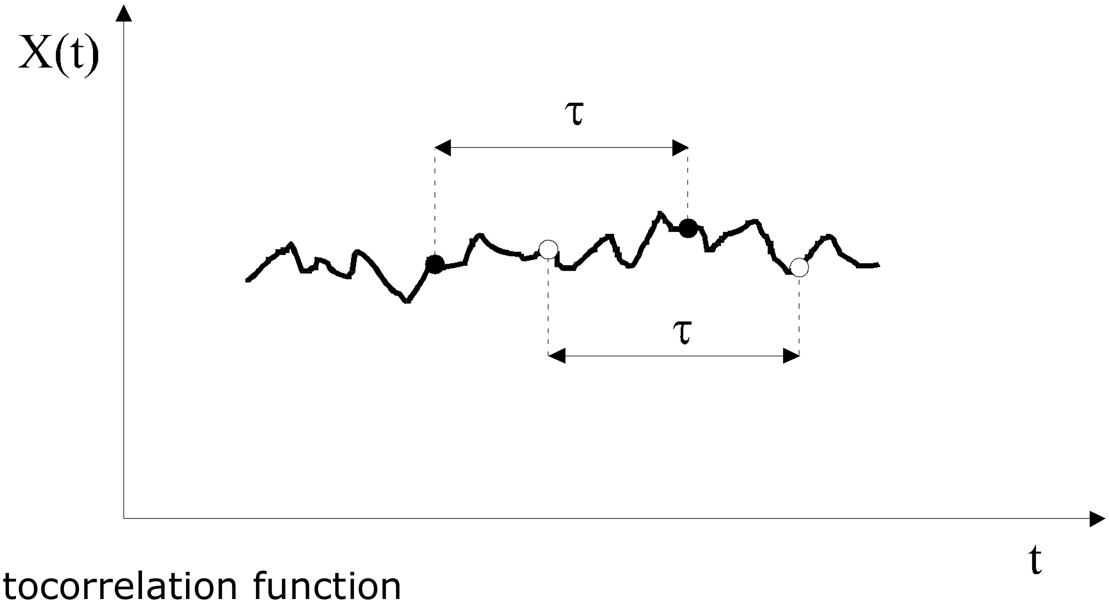
Exercise 3

Application of autocorrelation function to signal analysis



Autocorrelation function

$$R_{xx}(\tau) = E\left[X(t)\cdot X(t+\tau)\right] = \overline{X(t)\cdot X(t+\tau)}$$

Autocovariance function

$$C_{xx}(\tau) = E\left[x(t)\cdot x(t+\tau)\right] = \overline{x(t)\cdot x(t+\tau)}$$

$$x(t) = X(t) - \overline{X}$$

Normalized autocovariance function

$$Q_{xx}(\tau) = \frac{C_{xx}(\tau)}{\sigma_X^2} \in \langle -1;1 \rangle$$

An estimator of autocovariance function

$$C_{xx}(\tau) = C_{xx}(i\cdot\Delta t) = \frac{1}{N-i}\sum_{j=1}^{j=N-i} x_j \cdot x_{j+i}$$

Compulsory tasks

a) perform a selection of the maximum correlation distance τ_{max} for different signal types (periodic, random, compound) and try to relate it to the signal time scales

b) conduct the stationarity test of the autocorrelation function for the selected compound signal (determine the signal duration ensuring smooth correlation function distribution); noise content (see the appendix) should be at least 50%

c) determine the $Q_{xx}(\tau)$ distributions for various signal types (purely deterministic and distorted by noise)

d) analyse the relationships between the amplitude of the signal and the maximum value of the correlation function for various deterministic signals and propose the analytical formulae; repeat the analysis for "disturbed" signal, i.e. with added noise; refer to the lecture on correlation analysis, slides No 8, 9

☞ *please distinguish the difference (diagnostic value) between dimensional and normalised correlation functions*

e) analyse the evolution of the $Q_{xx}(\tau)$ distribution for a selected compound signal for varying noise content (see appendix); analyse the relationship between standard deviations (variances) of signal components and characteristic levels of the correlation function; refer to the lecture on correlation analysis, slide No 11

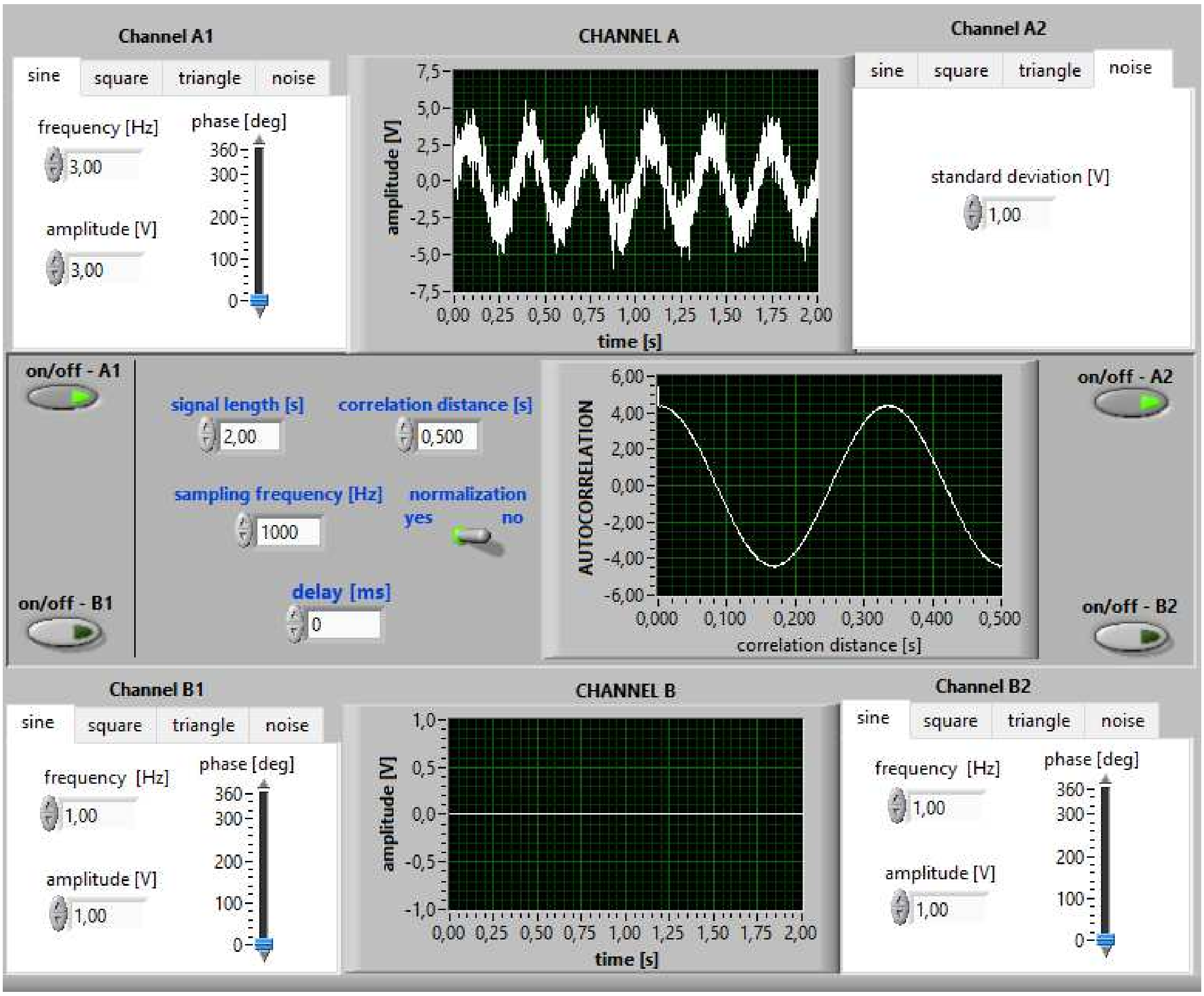
Optional tasks

- check whether the autocorrelation function has an even or odd distribution
- discuss the influence of the signal's phase angle on autocorrelation function distribution
- ....

☞ **The exercise aims at learning how to:**

- **accurately perform correlation analysis (choice of signal duration and max. delay)**
- **recover the information about the signal (type, composition, parameters) having known correlation function distribution (and, of course, not seeing the signal).**

project "Correlation" in LabView environment



Sample problems

1. What should be the relationship between max. correlation distance and signal duration ensuring the accurate estimation of correlation function?
2. What is the unit of the autocorrelation function and what is the range of values it may take ?
3. What is the unit of the normalized autocovariance function and what is the range of values it may take?
4. When the correlation function is periodic ?
5. What value of max. correlation distance would you choose for the correlation analysis of a sine wave oscillating with the frequency of 50Hz. Please explain your answer.
6. Describe (by drawing or in written form) the distribution of the autocorrelation function of the selected compound signal. Indicate the relationships between signal parameters and correlation function.
7. What is the value of correlation distance corresponding to the maximum of the autocovariance function ? What is the maximum C_{xx} value ?
8. What are the differences between autocorrelation function distributions corresponding to the following sine waves:

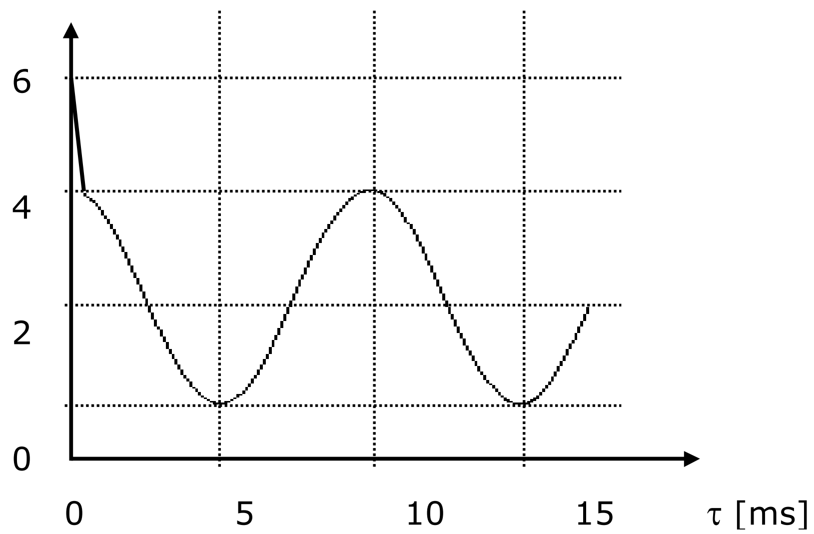
   a) $y_1 = \mathbf{3} + A \cdot \sin(2\pi f \cdot t + \varphi)$
   $y_2 = \mathbf{5} + A \cdot \sin(2\pi f \cdot t + \varphi)$

   b) $y_1 = y_{sr} + \mathbf{2} \cdot A \cdot \sin(2\pi f \cdot t + \varphi)$
   $y_2 = y_{sr} + \mathbf{4} \cdot A \cdot \sin(2\pi f \cdot t + \varphi)$

   c) $y_1 = y_{sr} + A \cdot \sin(2\pi \cdot \mathbf{f} \cdot t + \varphi)$
   $y_2 = y_{sr} + A \cdot \sin(2\pi \cdot \mathbf{(2 \cdot f)} \cdot t + \varphi)$

   d) $y_1 = y_{sr} + A \cdot \sin(2\pi f \cdot t + \varphi)$
   $y_2 = y_{sr} + A \cdot \sin(2\pi f \cdot t + \mathbf{(\varphi + \pi)})$

   Compare R_{xx} distributions graphically.
9. Having known the distribution of the autocorrelation function describe the signal, i.e. provide its composition and parameters.



Appendix

Let's consider the signal X(t). Its power is given as

$$N = \lim_{T\to\infty}\frac{1}{T}\int_0^T X^2(t)dt$$

When the signal composition is known, then the power can be expressed as a sum of partial powers N_i

$$N = \sum_i N_i$$

For instance, if the signal is composed of a sine wave with a non-zero mean value $\overline{X}$ and disturbed by a noise then

$$N = N_{sin} + N_{noise} + N_{mean} = \sigma_{sin}^2 + \sigma_{noise}^2 + \overline{X}^2 = \frac{A^2}{2} + \sigma_{noise}^2 + \overline{X}^2$$

where

σ_{sin} - standard deviation of the sine wave
A - amplitude of the sine wave
σ_{noise} - standard deviation of the noise

The content of the noise in the signal can be calculated as follows

$$\varepsilon = \frac{N_{noise}}{N - N_{mean}} \cdot 100\%$$